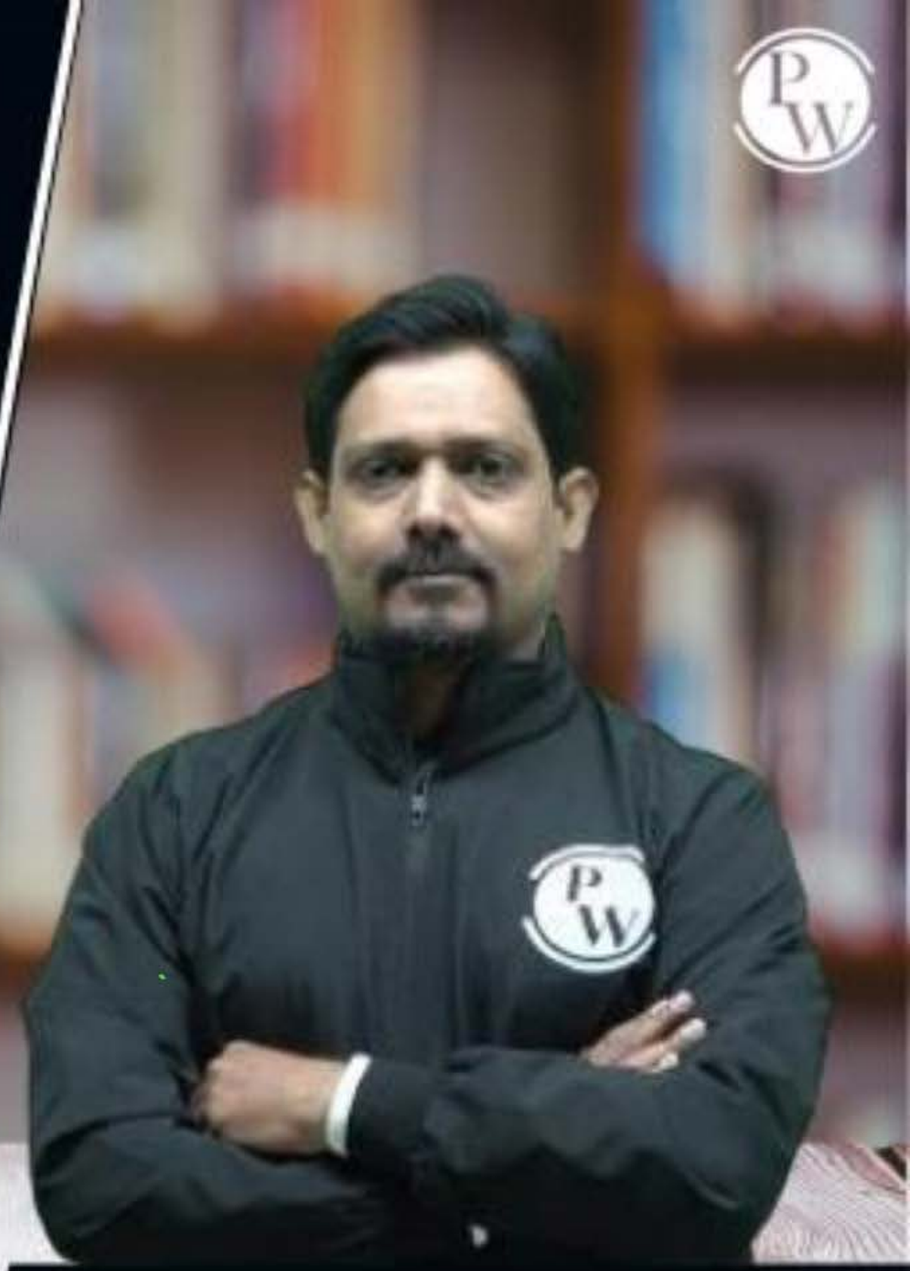
Computer Science & IT



Data Structure & Programming

Graph & Hashing

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



Topic

Topic

Topic

Topic

Topic

Graph BFS

Hashing

Chaining

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Open Hashing

Closed Hashing / Open Addressing

Linear probing

Quadratic probing

Double Hashing

Slide



Topic : Hashing

Chaining

3rd Insertion $\frac{97}{100}$

Q. Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform hashing, what is the probability that the first 3 slots are unfilled after the first three insertions?

(A) $(97 \times 97 \times 97)/100^3$

(B) $(99 \times 98 \times 97)/100^3$

(C) $(97 \times 96 \times 95)/100^3$

(D) $(97 \times 96 \times 95)/(3! \times 100^3)$



first Insertion

$\frac{\text{favourable cases}}{\text{No. of cases}} = \frac{97}{100} \times$

2nd Insertion

$= \frac{97}{100}$

Slide



Topic : Hashing

In chaining Slot is Not blocked

because we donot store within the table.

Slide



Topic : Hashing

Closed Hashing : In case of collision we store the within the table in different position.

1. Linear probing
2. Quadratic probing
3. Double Hashing

} Closed Hashing also call

Open Addressing.

Slide



Topic : Hashing

wrap around

Linear probing

If collision occurs the the next empty position

Search in linear hashing in the table

Slide



Topic : Hashing

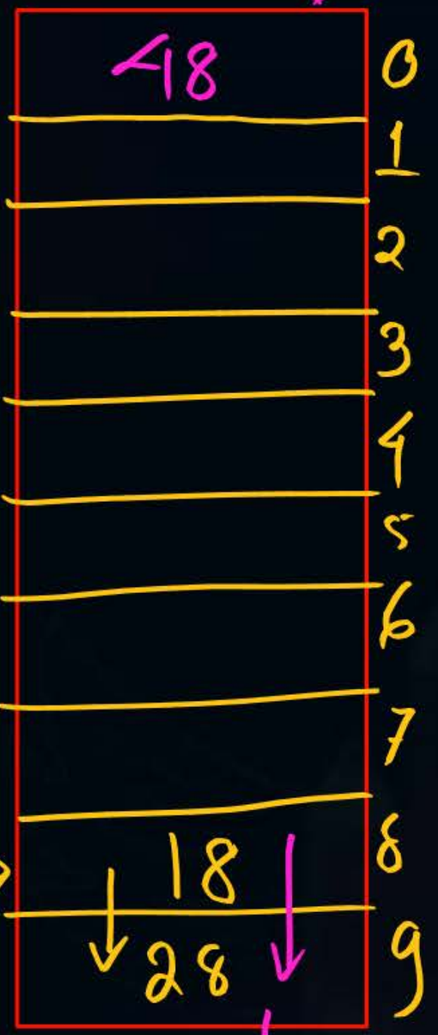
Linear probing

$h(k) = k \bmod (10)$

$h(18) = 18 \bmod 10 = 8$

$h(28) = 28 \bmod 10 = 8$

$h(48) = 48 \bmod 10 = 8$



Circular manner

Next empty position searched

Slide

Linear Probing Example

Consider the the hash table of size 13, the hash table is initially empty and hash $h(k) = k \ mod \ 13$ is used. Collision is resolved by *linear probing*.

Insert keys:

18, 41, 22, 44, 59, 32, 31, 73

in to hash table and show the resultant hash table.

18 % 13 = 5

41 % 13 = 2

22 % 13 = 9

44 % 13 = 5

59 % 13 = 7

32 % 13 = 6

31 % 13 = 5

73 % 13 = 8



Slide



Topic : Hashing

A hash table contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is key % 10. if the values 43, 165, 62, 123, 142 are inserted in the table, in what location would the key value 142 be inserted?

(A) 2

(B) 3

(C) 4

(D) 6



Slide



Topic : Hashing

A hash table of length 10 uses open addressing with hash function h(k) = kmod10, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

Which one of the following choices gives a possible order in which the key values could have been inserted in the table?

A) 46, 42, 34, 52, 23, 33

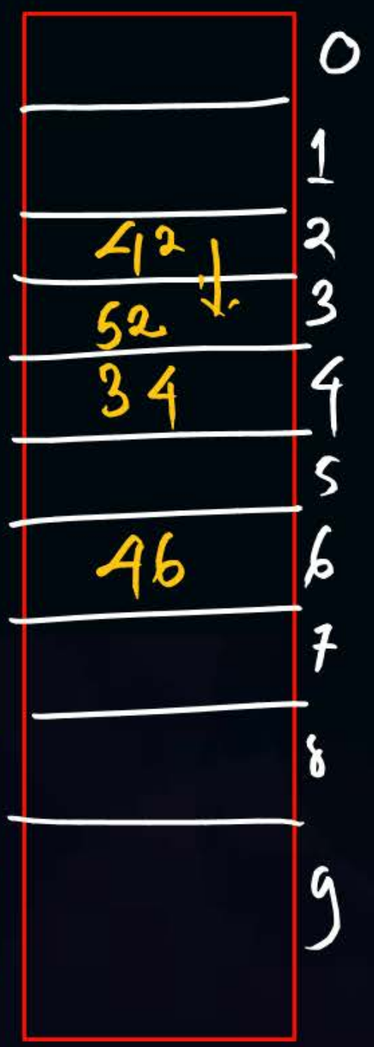
B) 34, 42, 23, 52, 33, 46

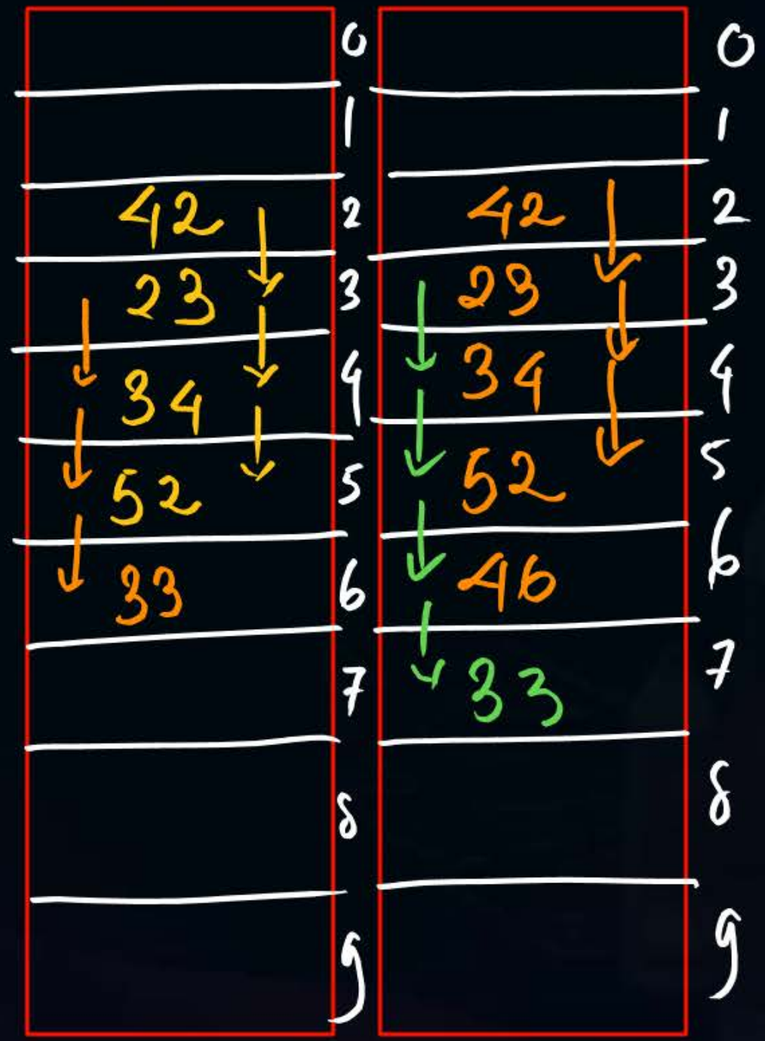
C) 46, 34, 42, 23, 52, 33

D) 42, 46, 33, 23, 34, 52

Slide

Topic : Hashing





0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

A) 46, 42, 34, 52, 23, 33

B) 34, 42, 23, 52, 33, 46

C) 46, 34, 42, 23, 52, 33

D) 42, 46, 33, 23, 34, 52

Slide



Topic : Hashing

How many different insertion sequences of the key values using the same hash function and linear probing will result in the hash table shown above?

(A) 10

(B) 20

(C) 30

(D) 40

Insertion Sequence

4 elements, 42, 23, 34, 46 (52, 33)

1 fixed

_ _ _ _

4!

= 24

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

Slide

Topic : Hashing

How many different insertion sequences of the key values using the same hash function and linear probing will result in the hash table shown above?

(A) 10

(B) 20

(C) 30

(D) 40

42, 23, 34, 52, 46, 33

3! = 6

2 4 + 6 = 30

0	
1	
2	42
3	23
4	34
5	52
6	
7	
8	
9	

Slide

Linear Probing Example

18, 41, 22, 44, 59, 32, 31, 73

Key	$h(x) = x \bmod 13$
18	5
41	2
22	9
44	5
59	7
32	6
31	5
73	8

	0
	1
41	2
	3
	4
18	5
44	6
59	7
32	8
22	9
31	10
73	11
	12

primary clustering

Slide



Topic : Hashing

Linear probing suffer from problem of [primary clustering]

* Once the cluster forms bigger it get faster it grows.
* Insert & search will take more time.

Hashing performance degrade.

Slide



Topic : Hashing

To overcome problem of primary cluster we use quadratic probing.

Slide



Topic : Hashing



$h(k)+1^2$ first

$h(k)+2^2$ Second collision

$h(k)+3^2$ Third collision

$h(k)+4^2$ fourth collision

$h(k)+5^2$ fifth collision

Slide

Topic : Hashing

96, 48, 63, 29, 87, 77, 47, 73, 70, 86, 166

$96 \bmod 20 = 16$

$48 \bmod 20 = 8$

$63 \% 20 = 3$

$29 \% 20 = 9$

$87 \% 20 = 7$

$77 \% 20 = 17$

$47 \% 20 = 7$ Collision.

1st probe $= (7 + 1^2) \% 20 = 8$

2nd probe $= (7 + 2^2) \% 20 = 11$

$73 \% 20 = 13$

$70 \% 20 = 10$

$86 \% 20 = 6$

$166 \% 20 = 6$

$(6 + 1^2) \% 20 = 7$

$(6 + 2^2) \% 20 = 10$

$(6 + 3^2) \% 20 = 15$



Slide

Example

{96, 48, 63, 29, 87, 77, 47, 73, 70, 86, 166}.

Example

{96, 48, 63, 29, 87, 77, 47, 73, 70, 86, 166}.

$Key - x$	$h(x,i) = (h(x), i^2)\ mod\ 20$
96	$i = 0, (h(96), 0) mod\ 20 = 16$
48	$i = 0, (h(48), 0) mod\ 20 = 8$
63	$i = 0, (h(63), 0) mod 20 = 3$
29	$i = 0, (h(29), 0) mod 20 = 9$
87	$i = 0, (h(87), 0) mod 20 = 7$
77	$i = 0, (h(77), 0) mod 20 = 17$
47	$i = 0, (h(47), 0) mod 20 = 7$
73	$i = 0, (h(73), 0) mod 20 = 13$
70	$i = 0, (h(70), 0) mod 20 = 10$
86	$i = 0, (h(86), 0) mod 20 = 6$
166	$i = 2, (h(166), 4) mod 20 = 16$

	0
	1
	2
63	3
	4
	5
86	6 Occupied
87	7 Occupied
48	8 Occupied
29	9
70	10 Occupied
47	11
	12
73	13
	14
166	15
96	16
77	17
	18
	19



2 mins Summary

Topic

Linear probing

Topic

quadratic probing

Topic

Topic

Topic

Slide

THANK - YOU